

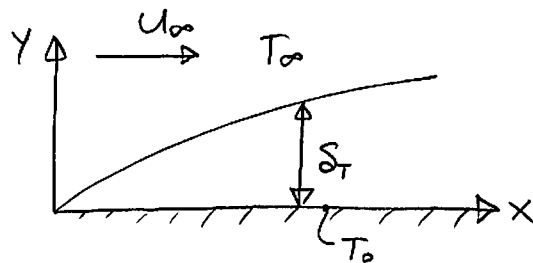
Heat Transfer (Similarity Solution)

$$h = \frac{q''|_{y=0}}{\Delta T} = \frac{q''|_{y=0}}{T_\infty - T_0} = ?$$

Looking at our energy equation:

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2}$$

$$\begin{aligned} T(y=0) &= T_0 \\ T(y \rightarrow \infty) &= T_\infty \end{aligned}$$



Non dimensionalizing: $\Theta = \frac{T - T_0}{T_\infty - T_0}$

$$u \frac{\partial \Theta}{\partial x} + v \frac{\partial \Theta}{\partial y} = \alpha \frac{\partial^2 \Theta}{\partial y^2} \Leftrightarrow u \frac{\partial \bar{u}}{\partial x} + v \frac{\partial \bar{u}}{\partial y} = \nu \frac{\partial^2 \bar{u}}{\partial y^2}$$

B.l. momentum equation

$$\Theta(y=0) = 0$$

$$\Theta(y \rightarrow \infty) = 1$$

$$\left. \frac{\partial \Theta}{\partial y} \right|_{y \rightarrow \infty} = 0$$

Noting that a usefull connection between the energy & hydrodynam. equations is $Pr = \nu/\alpha$

If $Pr = 1$, $\alpha = \nu$ and $\Theta = \bar{u}$ and $\delta = \delta_T$

Usually however, $Pr \neq 1$, so we have to solve for Θ

Assuming an identical similarity variable as before

$$\eta = y \sqrt{\frac{U_\infty}{x\nu}}$$

$$\frac{\partial \Theta}{\partial x} = \frac{\partial \Theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial x} = \frac{\partial \Theta}{\partial \eta} \cdot \left(-\frac{y}{2} \sqrt{\frac{U_\infty'}{\nu x^3}} \right)$$

$$\frac{\partial \theta}{\partial y} = \frac{\partial \theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial y}$$

$$\begin{aligned} \frac{\partial^2 \theta}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{\partial \theta}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{\partial \theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{\partial \theta}{\partial \eta} \right) \cdot \left(\frac{\partial \eta}{\partial y} \right) + \frac{\partial}{\partial y} \left(\frac{\partial \eta}{\partial y} \right) \left(\frac{\partial \theta}{\partial \eta} \right) \\ &= \frac{\partial}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} \left(\frac{\partial \theta}{\partial \eta} \right) \cdot \frac{\partial \eta}{\partial y} = \frac{\partial^2 \theta}{\partial \eta^2} \left(\frac{\partial \eta}{\partial y} \right)^2 \end{aligned}$$

Back substituting into our energy equation:

$$u \frac{\partial \theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial x} + v \frac{\partial \theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} = \alpha \frac{\partial^2 \theta}{\partial \eta^2} \cdot \left(\frac{\partial \eta}{\partial y} \right)^2 \quad (1)$$

To solve, we need to use our hydrodynamic b.l. definitions:
On page 48 of notes, we already solved that:

$$\frac{\partial \eta}{\partial x} = -\frac{1}{2x} \cdot \eta$$

$$\frac{\partial \eta}{\partial y} = \sqrt{\frac{U_\infty}{x\nu}}$$

$$u = U_\infty f' \quad (\text{Definition, we defined } f' = \frac{u}{U_\infty})$$

$$v = U_\infty \cdot \frac{1}{2x} \sqrt{\frac{x\nu}{U_\infty}} (\eta f' - f) \quad (\text{From continuity})$$

Back substituting into equation (1)

$$\begin{aligned} -U_\infty f' \frac{\partial \theta}{\partial \eta} \cdot \underbrace{\frac{y}{2x} \sqrt{\frac{U_\infty}{x\nu}}}_{\frac{\eta}{2}} + \frac{U_\infty}{2x} \sqrt{\frac{x\nu}{U_\infty}} (\eta f' - f) \frac{\partial \theta}{\partial \eta} \sqrt{\frac{U_\infty}{x\nu}} \\ = \alpha \frac{\partial^2 \theta}{\partial \eta^2} \cdot \frac{U_\infty}{x\nu} \end{aligned}$$

$$\begin{aligned} -f' \frac{\partial \theta}{\partial \eta} \cdot \frac{\eta}{2} + f' \frac{\partial \theta}{\partial \eta} \cdot \frac{\eta}{2} \cdot \cancel{\sqrt{\frac{x\nu}{U_\infty}}} \cdot \cancel{\sqrt{\frac{U_\infty}{x\nu}}} - \frac{1}{2} f \frac{\partial \theta}{\partial \eta} \cancel{\sqrt{\frac{x\nu}{U_\infty}}} \cdot \cancel{\sqrt{\frac{U_\infty}{x\nu}}} \\ = \alpha \frac{\partial^2 \theta}{\partial \eta^2} \frac{1}{\nu} \end{aligned}$$

The first two terms cancel and we are left with:

$$\frac{\partial^2 \theta}{\partial \eta^2} + \frac{1}{2} \underbrace{\frac{\nu}{\alpha}}_{Pr} f \frac{\partial \theta}{\partial \eta} = 0$$

$$\boxed{\theta'' + \frac{1}{2} Pr \cdot f \cdot \theta' = 0} \quad (2)$$

We've turned our energy PDE into an ODE
To solve we can integrate but here we can use a trick:
Usually most books do the following:

$$\frac{d\theta'}{d\eta} + \frac{1}{2} Pr \cdot f \cdot \theta' = 0$$

$$\frac{d\theta'}{\theta'} + \frac{Pr}{2} \cdot f \cdot d\eta = 0$$

$$\theta = C_1 \int_0^\eta \left[\exp\left(-\frac{Pr}{2} \int_0^\eta f d\eta\right) \right] d\eta + C_2$$

Using our b.c.'s $\Rightarrow \theta = 0$ at $\eta = 0 \Rightarrow C_2 = 0$

$$\theta = 1 \text{ at } \eta \rightarrow \infty \Rightarrow C_1 = \frac{1}{\int_0^\infty \left[\exp\left(-\frac{Pr}{2} \int_0^\eta f d\eta\right) \right] d\eta}$$

So our cumbersome solution is:

$$\boxed{\theta(\eta) = \frac{\int_0^\eta \left[\exp\left(-\frac{Pr}{2} \int_0^\eta f d\eta\right) \right] d\eta}{\int_0^\infty \left[\exp\left(-\frac{Pr}{2} \int_0^\eta f d\eta\right) \right] d\eta}}$$

③ $\Rightarrow f(\eta)$ is tabulated from the Blasius solution, so we can solve this numerically.

However, there is an easier way to solve this:

Let:

$$f(\eta^*) = Pr^{2/3} \cdot f(\eta) \quad (4)$$

$$\text{and } \eta^* = \eta \cdot Pr^{1/3} \quad (5)$$

Back substitute (4) and (5) into (2)

$$\theta'' + \frac{1}{2} f(\eta^*) Pr^{1/3} \frac{\partial \theta}{\partial \eta} = 0$$

From our definition (eq. 5) $\Rightarrow \eta^* = \eta Pr^{1/3}$
 $d\eta^* = Pr^{1/3} d\eta$

$$\theta'' + \frac{1}{2} f(\eta^*) Pr^{2/3} \underbrace{\frac{\partial \theta}{\partial \eta Pr^{1/3}}}_{\partial \eta^*} = 0 \quad (6)$$

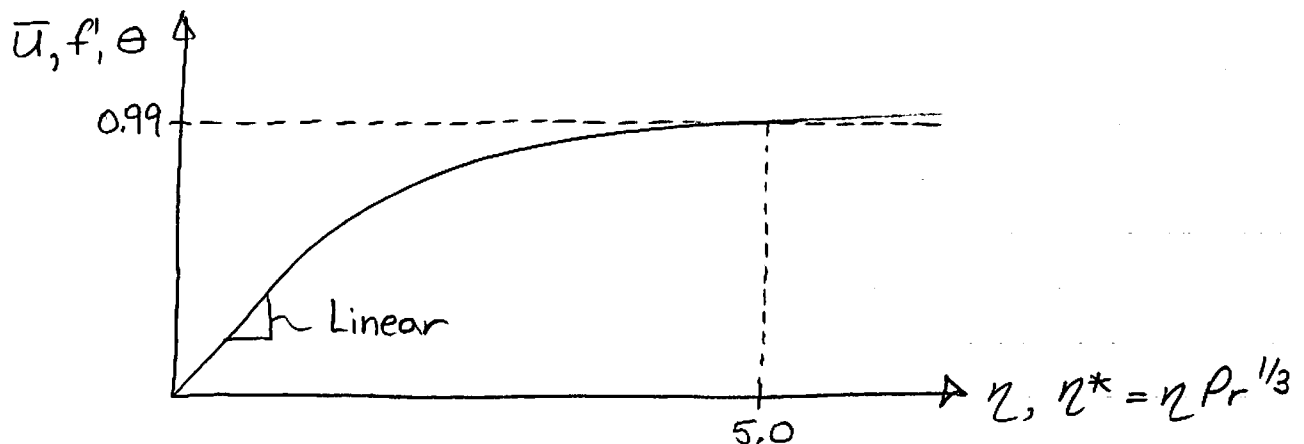
Multiply ⑥ through by $Pr^{-2/3}$

$$\underbrace{\frac{1}{Pr^{2/3}} \left(\frac{\partial^2 \theta}{\partial \eta^2} \right)}_{\partial^2 \eta^{*2}} + \frac{1}{2} f(\eta^*) Pr^{2/3} \frac{\partial \theta}{\partial \eta^*} = 0$$

Now our PDE becomes identical to the Blasius ODE. We already have the solution.

$$\theta'' + \frac{1}{2} f(\eta^*) \theta' = 0 \Rightarrow \theta(\eta^*) \Leftrightarrow \bar{u}(\eta) = f'$$

Our b.c.'s are: $\left. \begin{array}{l} \theta(\eta^*=0) = 0 \\ \theta(\eta^* \rightarrow \infty) = 1 \end{array} \right\} \theta = \frac{T - T_0}{T_\infty - T_0}$



$\eta \equiv$ hydrodynamic b.l.
 $\eta^* \equiv$ thermal b.l.

For the hydrodynamic boundary layer, we had:

$$\delta \sqrt{\frac{U_\infty}{x\nu}} = \eta(y=\delta) = 5.0 \Rightarrow \boxed{\delta(x) = \frac{5x}{\sqrt{Re_x}}}$$

Now for the thermal boundary layer, our solution is similar

$$Pr^{1/3} \delta_T \sqrt{\frac{U_\infty}{x\nu}} = \eta^*(y=\delta_T) = 5.0 \Rightarrow \boxed{\delta_T(x) = \frac{5x}{Re_x^{1/2} Pr^{1/3}}}$$

Taking the ratio of our two boundary layer thicknesses:

$$\boxed{\frac{\delta}{\delta_T} = Pr^{1/3}} \Rightarrow \text{Kind of intuitive since } Pr = \frac{\nu}{\alpha} = \frac{\text{hydrodynamics}}{\text{energy}}$$

Note before on page (45) of notes, we found with scaling that:

$$\delta_T \sim L Pr^{-1/3} Re_L^{-1/2} \quad \left. \vphantom{\delta_T} \right\} \text{ Here we have } \delta_T(L) = 5L Re_L^{-1/2} Pr^{-1/3}$$

Now we can solve for our heat transfer (note we know: $h \sim \frac{k}{\delta_T}$)

$$q''|_{y=0} = -k \frac{\partial T}{\partial y}|_{y=0} \Rightarrow \Theta = \frac{T-T_0}{T_\infty-T_0} ; \quad \partial \Theta = \frac{\partial T}{T_\infty-T_0}$$

$$\eta^* = \eta Pr^{1/3} = y \left(\frac{U_\infty}{x\nu} \right)^{1/2} Pr^{1/3}$$

$$\partial \eta^* = \partial y \left(\frac{U_\infty}{x\nu} \right)^{1/2} Pr^{1/3}$$

$$q''|_{y=0} = -k (T_\infty - T_0) \left(\frac{U_\infty}{x\nu} \right)^{1/2} Pr^{1/3} \frac{\partial \Theta}{\partial \eta^*} \Big|_{\eta^*=0}$$

$$= + \frac{k (T_0 - T_\infty)}{x} \left(\frac{U_\infty x}{\nu} \right)^{1/2} Pr^{1/3} f''(0)$$

Remember we already know $f \Rightarrow f = \frac{a_2 \eta^2}{2!} - \frac{a_2 \eta^5}{2 \cdot 5!} + \frac{11}{4} \frac{a_3 \eta^8}{8!} \dots$
From page (50) of notes

$$f''(0) = a_2 = 0.332$$

$$q''|_{y=0} = \frac{k\Delta T}{x} Re_x^{1/2} Pr^{1/3} a_2$$

$$Nu_x = \frac{hx}{k} = \underbrace{\frac{q''|_{y=0}}{\Delta T}}_h \cdot \frac{x}{k} = a_2 Re_x^{1/2} Pr^{1/3}$$

$$Nu_x = 0.332 Re_x^{1/2} Pr^{1/3}$$

⇒ From scaling: $Nu_L \sim Re_L^{1/2} Pr^{1/3}$
 ⇒ Valid for $Pr > 0.5$, $T_o = \text{constant}$

Colburn Analogy

We can relate our fluid flow & heat transfer solutions

$$C_{f,x} = \frac{\tau}{\frac{1}{2}\rho U_\infty^2} = \frac{2a_2}{Re_x^{1/2}} \quad (\text{skin friction coefficient})$$

$$Nu_x = a_2 Re_x^{1/2} Pr^{1/3}$$

Let's try the following

$$\frac{Nu_x}{Re_x \cdot Pr} = \frac{a_2 Re_x^{1/2} Pr^{1/3}}{Re_x \cdot Pr} = \frac{a_2}{Re_x^{1/2} Pr^{2/3}} = \frac{1}{2} \left(\frac{2a_2}{Re_x^{1/2}} \right) \frac{1}{Pr^{2/3}} \overset{C_{f,x}}{=}$$

So what's the point of dividing by Pr ?

Let's expand our definition:

$$\frac{Nu_x}{Re_x \cdot Pr} = \frac{hx}{k} \cdot \frac{\mu}{\rho U_\infty x} \cdot \frac{\alpha}{U} = \frac{h}{k} \cdot \frac{\mu}{\rho U_\infty} \cdot \frac{\alpha}{U} \overset{\alpha}{=} \frac{h}{\rho C_p U_\infty} \overset{U}{=} \frac{h}{\rho C_p U_\infty}$$

$$\frac{Nu_x}{Re_x \cdot Pr} = \frac{h}{\rho C_p U_\infty} = St \equiv \text{Stanton Number}$$

$$St = \frac{\text{heat transfer to fluid}}{\text{thermal capacity of fluid}} \sim \frac{T_w - T_b}{T_{b,t=2} - T_{b,t=1}}$$

We can now say the following:

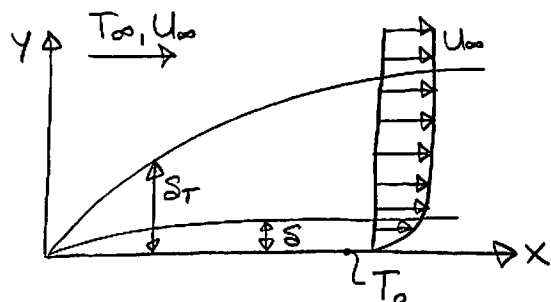
$$\boxed{St \cdot Pr^{2/3} = \frac{C_{f,x}}{2} = j_H \equiv \text{Colburn } j\text{-factor}} \quad (\text{or Colburn-Chilton analogy})$$

Very useful for heat transfer analysis as it relates transport properties to one another, (heat, momentum, and mass transfer).

Note the previous solution is valid for $Pr \approx 1$, or $Pr \gg 1$. What if $Pr \ll 1$.

Think about why this changes:

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2}$$



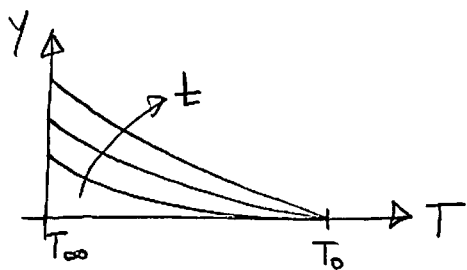
If $Pr \ll 1$, $u \propto Re_x^{-1/2}$ since $\delta \ll \delta_T$. Hence $u \sim U_\infty$. Our energy equation becomes:

$$U_\infty \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2}$$

From continuity: $\frac{\partial u}{\partial x} = -\frac{\partial v}{\partial y} \Rightarrow u = U_\infty = \text{constant} \Rightarrow v = \text{const.} = 0$ outside the b.l. $\downarrow$

$$U_\infty \frac{\partial T}{\partial x} = \alpha \frac{\partial^2 T}{\partial y^2} \Rightarrow \text{but } x = U_\infty \cdot t \Rightarrow dx = U_\infty dt$$

$$U_\infty \frac{\partial T}{U_\infty dt} = \alpha \frac{\partial^2 T}{\partial y^2} \Rightarrow \boxed{\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial y^2}} \Rightarrow \text{Transient conduction!!!}$$



$$q''_{y=0} = \frac{k \Delta T}{(\pi \alpha t)^{1/2}} \Rightarrow \text{Solved in ME420 Look up in conduction}$$

$$Nu_x = \left(\frac{q''_{y=0}}{\Delta T} \cdot \frac{x}{k} \right) = \frac{x}{(\pi \alpha t)^{1/2}} ; t = \frac{x}{U_\infty}$$

$$\boxed{Nu_x = \frac{1}{\pi^{1/2}} \cdot Re_x^{1/2} Pr^{1/2}} \Rightarrow Pr < 0.5, T_0 = \text{constant}$$